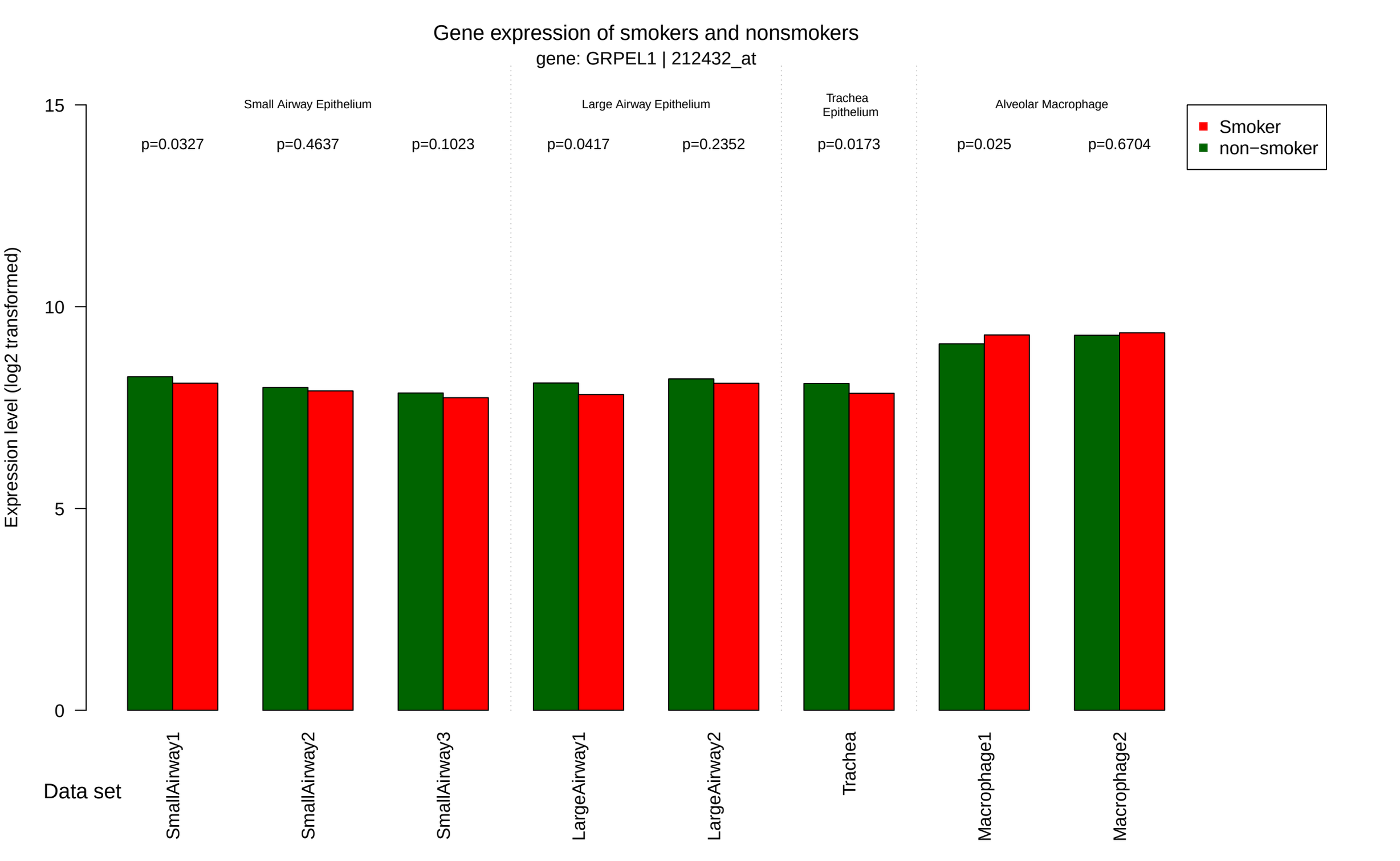
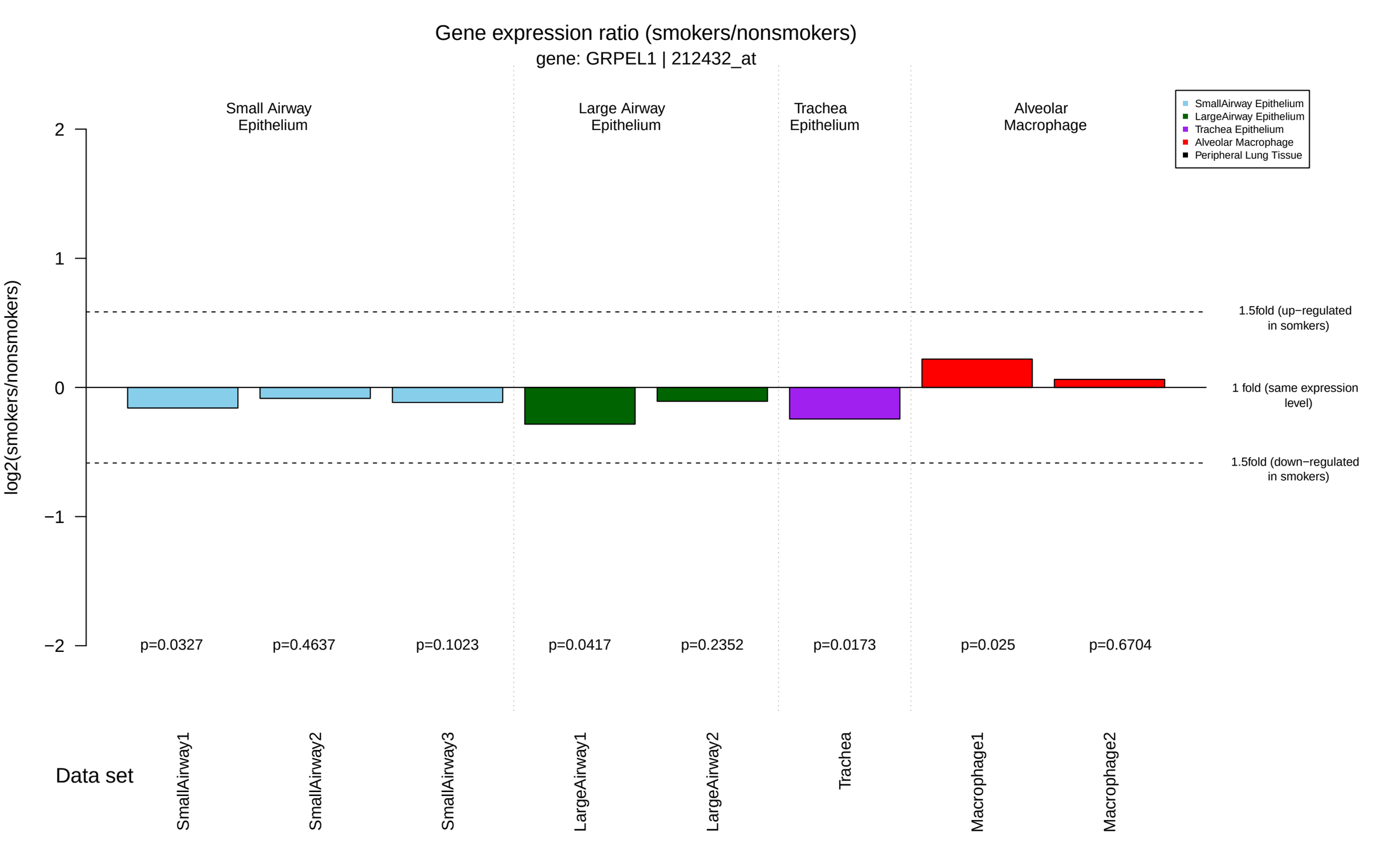
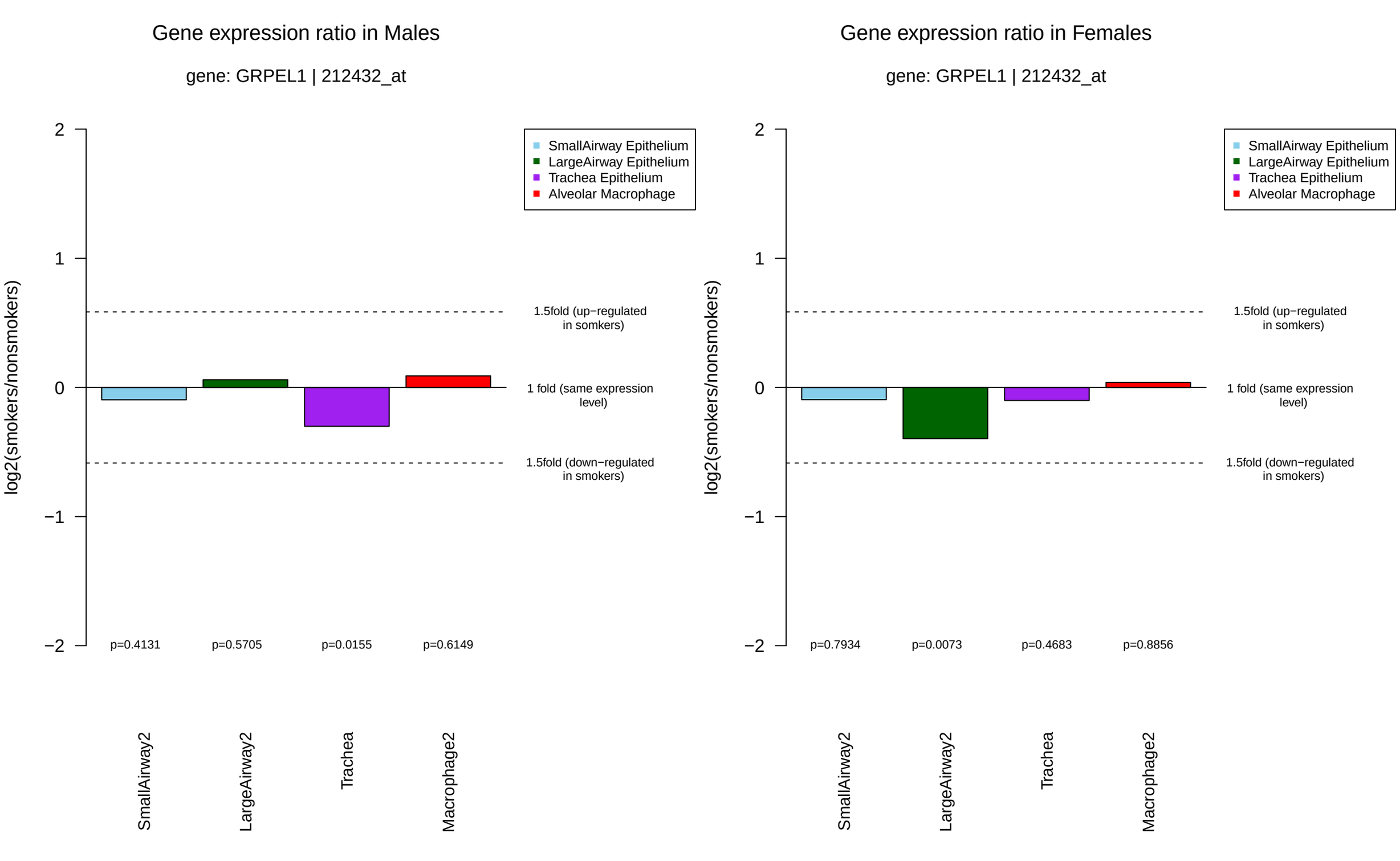
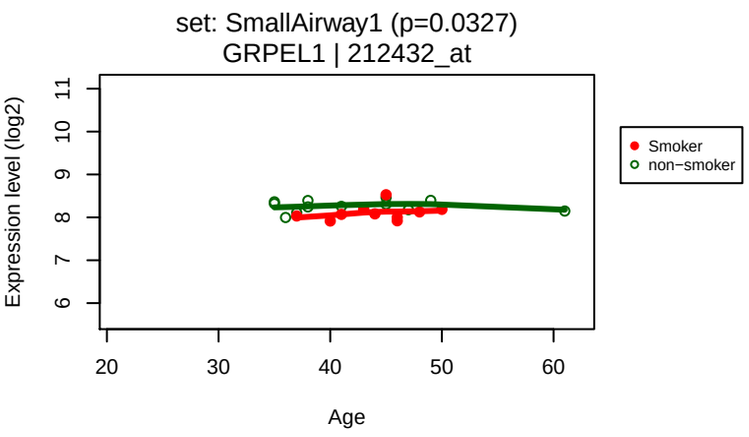
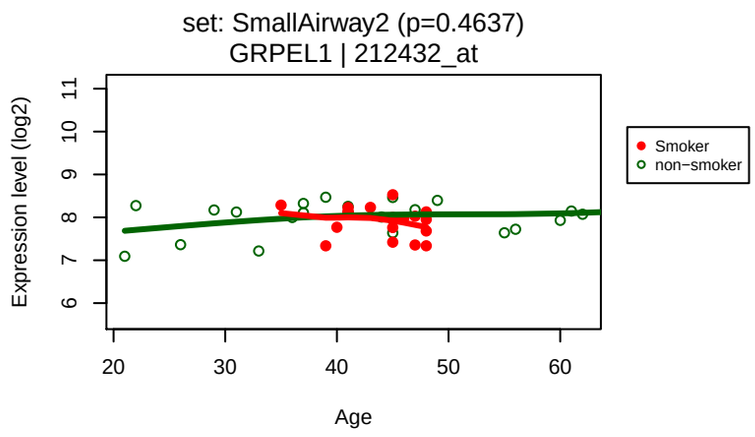
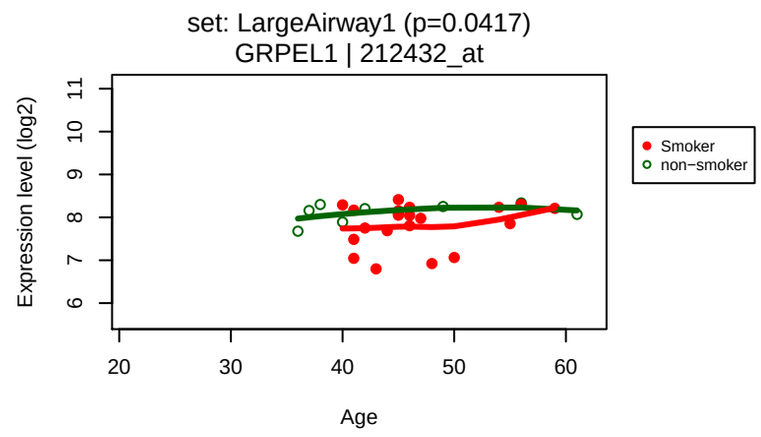

Gene expression of smokers and nonsmokers
gene: GRPEL1 | 212432_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0327
p=0.4637
p=0.1023
p=0.0417
p=0.2352
p=0.0173
p=0.025
p=0.6704
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

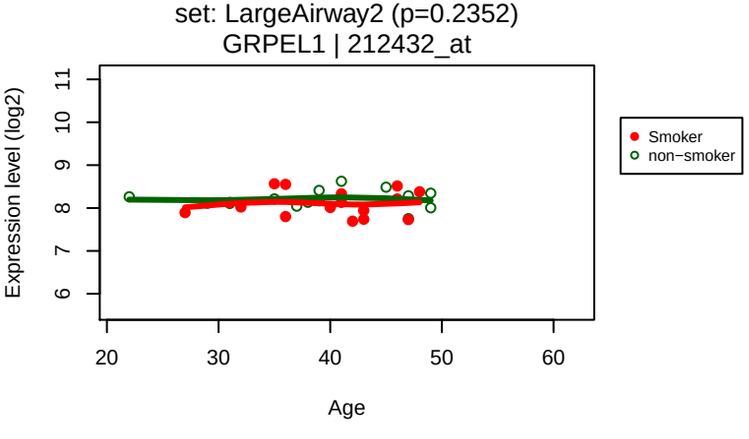

Gene expression ratio (smokers/nonsmokers)
gene: GRPEL1 | 212432_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0327
p=0.4637
p=0.1023
p=0.0417
p=0.2352
p=0.0173
p=0.025
p=0.6704
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

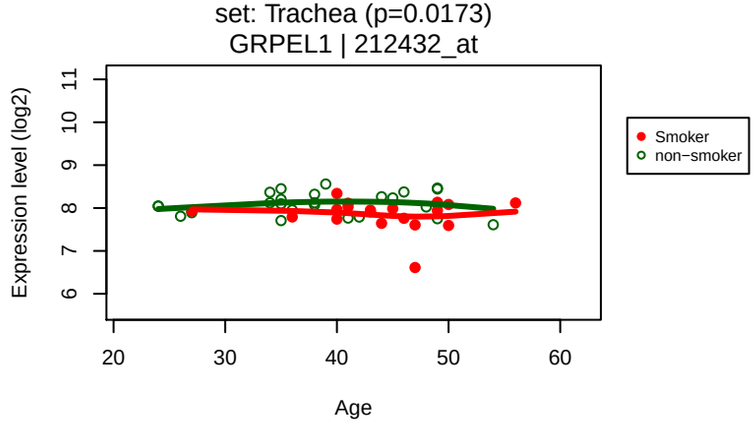

Gene expression ratio in Males
gene: GRPEL1 | 212432_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4131
p=0.5705
p=0.0155
p=0.6149
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GRPEL1 | 212432_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7934
p=0.0073
p=0.4683
p=0.8856
SmallAirway2
LargeAirway2
Trachea
Macrophage2

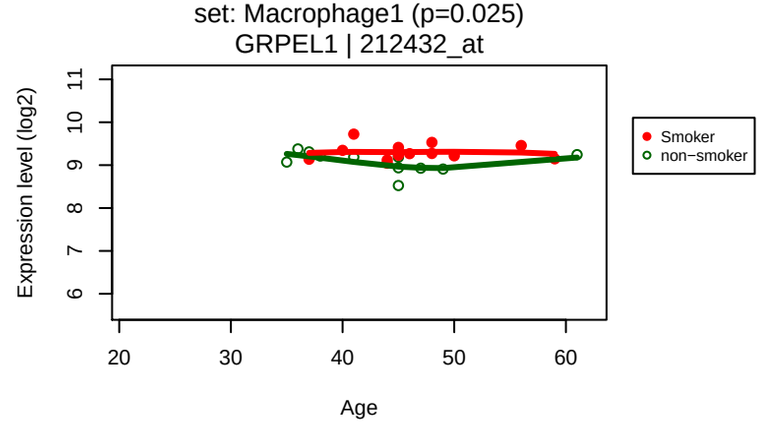

set: SmallAirway1 (p=0.0327)
GRPEL1 | 212432_at
Expression level (log2)
Age
Smoker
non−smoker

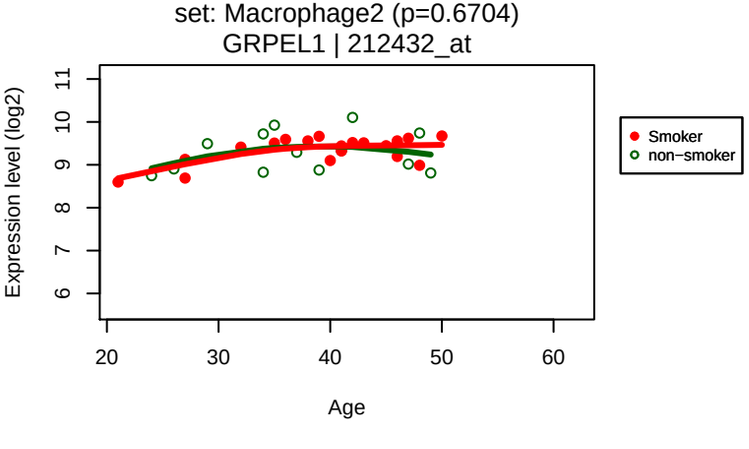

set: SmallAirway2 (p=0.4637)
GRPEL1 | 212432_at
Expression level (log2)
Age
Smoker
non−smoker

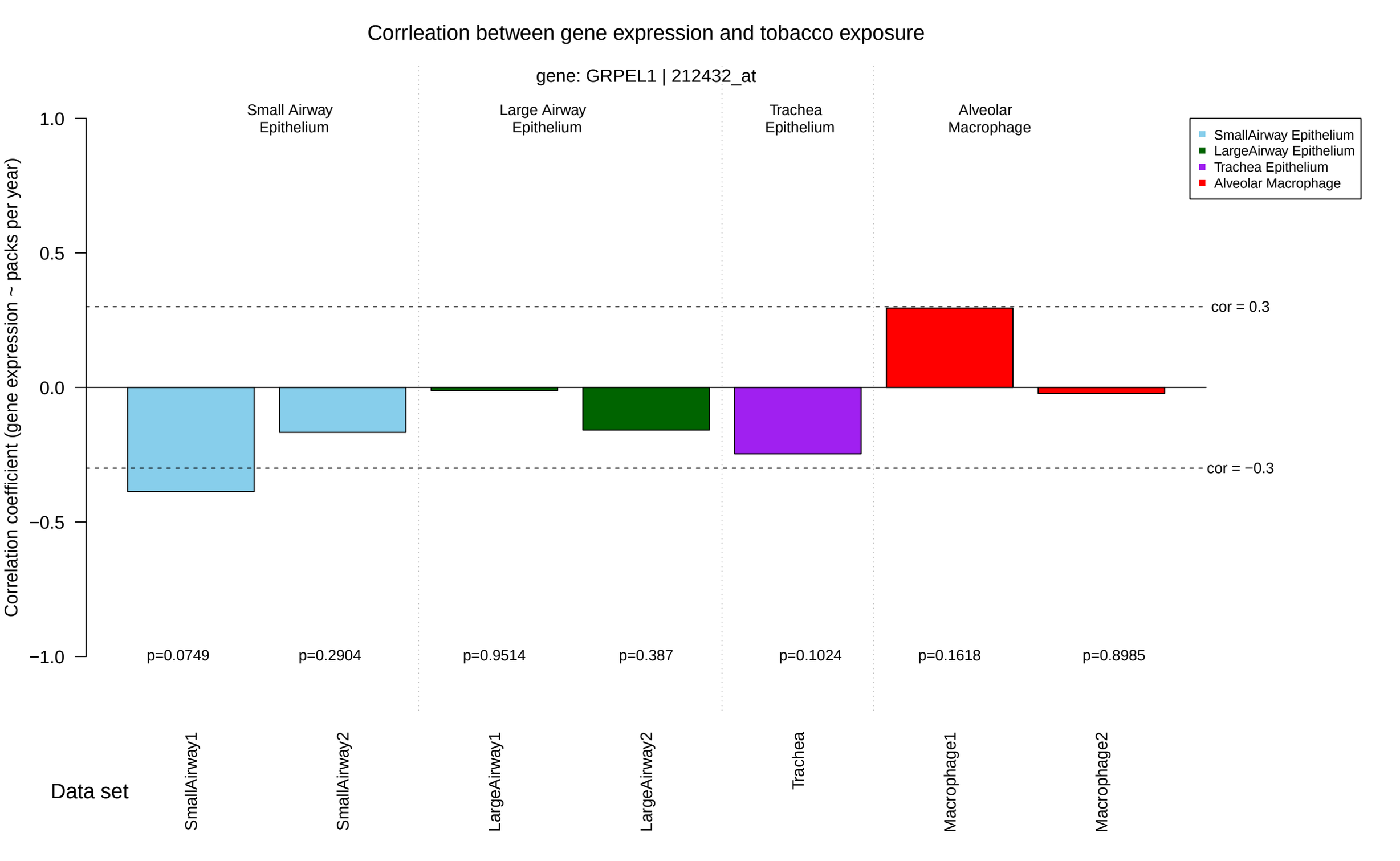

set: LargeAirway1 (p=0.0417)
GRPEL1 | 212432_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2352)
GRPEL1 | 212432_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0173)
GRPEL1 | 212432_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.025)
GRPEL1 | 212432_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6704)
GRPEL1 | 212432_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GRPEL1 | 212432_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0749
p=0.2904
p=0.9514
p=0.387
p=0.1024
p=0.1618
p=0.8985
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2